Where would a mixed-use building help?

Task 2.1 / Neighborhood evidence

Name _____ Team _____ Date _____

Design for someone who lives in the building and needs to reach a nearby business or service. Start with a place you know. A public intersection or school area works.

01 Read the neighborhood

In the app, open Peter Park's "Redesigning America," printed pp. 83-87. Find one planning choice that changes a resident's trip. Record the page and explain the effect in your words.

02 Walk through a real trip

Name a public starting point, a destination and a way to get there. What helps someone reach it? Where would a sidewalk, crossing, entrance or closer service make a difference?

03 Consider two responses to sprawl

Sprawl spreads homes and destinations farther apart. Describe one local effect. Compare two ways a change to the neighborhood could help.

Possible change	Who benefits and how?

Proposed site and the resident or service user you will design for

How much space does the design need?

Task 2.1 / Field measurements

Name _____ Team _____ Date _____

04 Calibrate a walking step

Mark a start point. Walk 10 ordinary steps in a straight, safe space. Your partner measures the distance in feet. Repeat three times. You can also direct a partner or use a tape for the room survey.

Walk	Distance for 10 steps (ft)	Distance ÷ 10 (ft/step)
1		
2		
3		
Average		

05 Survey a room

Use the same walker and count steps along two sides. Multiply each count by average ft/step. For a direct tape survey, enter the measured feet and mark the counts N/A.

Room and side	Steps	Estimated ft	Tape check (ft)	Difference (ft)
Length				
Width				

Area = length × width. Show both dimensions with units and calculate area in ft².

How many people usually use this room? Divide ft² by people. Name a second space that would need more or less room per person and explain why.

Draw a route a resident can actually use

Task 2.1 / Design brief and first plan

Name _____ Team _____ Date _____

06 Set the brief

Include housing access and one business or community service. Describe observable signs of success and real limits on space, materials, energy use or access.

Success criterion	Constraint or limit

07 Draw the ground floor

Label the street, both uses, entrances and one dimension with units. Trace one trip from a dwelling entrance to the service. Mark any point where that route leaves the building.

What feature of your plan responds to the need on sheet 1?

Test the trip and revise the design

Task 2.1 / Partner check and individual exit

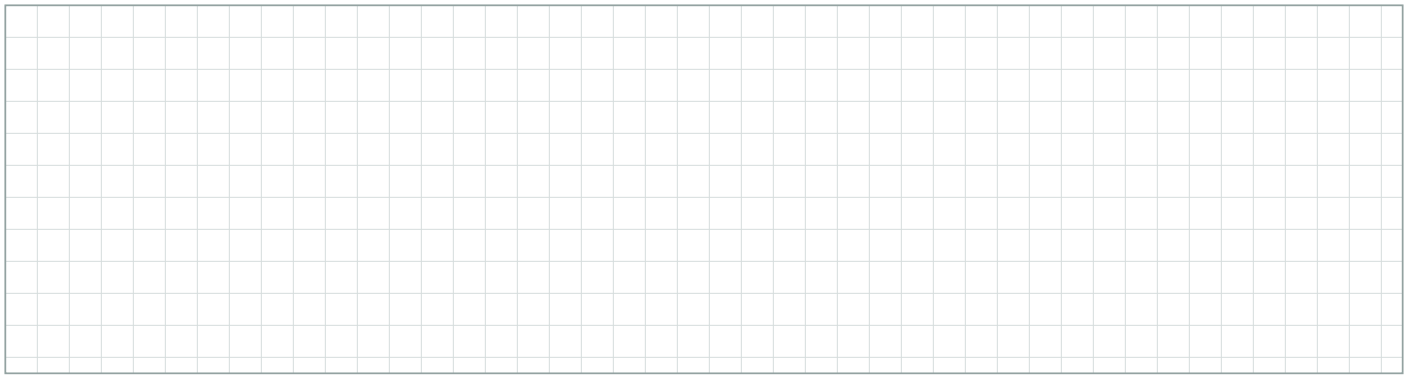
Name _____ Team _____ Date _____

08 Give the plan to another reader

Ask a partner to trace the trip without hints. Record their starting point, their destination and the first place they paused or found a barrier.

09 Revise one part of the route

Draw the revised entrance or route below. Label what moved. Add an arrow or dimension showing why the revision makes the trip work better.



Which criterion or constraint guided your revision?

10 Individual explanation

What local need does your building meet? Give a room dimension or area with units. Use one idea from Peter Park and one detail from your sketch to justify a design decision.

Turn in these sheets as directed. In the online lesson, enter your individual explanation under Explain. Complete the assigned reading check in the interactive reader.

Follow a load to the ground

Task 2.2 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Example building: apartments over a shop. Start at a person upstairs. Show how a floor, beam, column or wall, and foundation transfer the load to the ground.

Count a model deck’s self-weight as a permanent load. Mark where you will add equal test masses.

Strip span	Strip self-weight or description	Added mass	Support layout
Fixed setup			

02 Plan it on paper

Draw arrows from one person or object through floor, beam, support, and foundation.

My prediction, chosen feature or design limit

Follow a load to the ground

Task 2.2 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Span two low supports with the strip. Add a light load first at the center, then near one support. Keep the span fixed. Record the change in bend at both positions. Separate the added live load from the model's own weight.

Load position	Reference height	Loaded height	Bend and units
No added load			
Center			
Near support			

04 Use the reading

Ch. 11 | Bridging the Future | printed pp. 91-99
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Trace one load on your building sketch to the ground. Which part carries it next, and why?

Keep these sheets with the work. Save the individual explanation in the lesson's Explain step.

Find the first movement in a tower

Task 2.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Agree on height, allowed stock, and a stop rule with your teacher. Draw the tower's base and bracing. Mark the joint you predict will move first.

Measure the base and tower height. Record the starting lean or displacement.

Build target	Limit and unit	As built	Partner check
Height			
Footprint			
Stock / tape			

02 Plan it on paper

Sketch the bracing. Predict which joint or member will move first and how you will see it.

My prediction, chosen feature or design limit

Find the first movement in a tower

Task 2.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Build a freestanding tower to the recorded height using the class material limit. Have a partner check dimensions before loading. Load in equal increments under teacher supervision. Stop at the first major lean or joint slip. Photograph or sketch that first failure.

Increment	Total load and unit	Lean / displacement	First change or failure
0			
1			
2			
3			
4			

04 Use the reading

Ch. 12 | Tower in the Sky | printed pp. 101-109

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Identify the first failure and the load or trial when it appeared. What one change follows from it?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Compare beams under the same load

Task 2.4–2.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch both supports, the span and a fixed ruler reference at the center. Deflection is the change between unloaded and loaded position.

Measure each sample’s span, width, and thickness. Record differences that affect the comparison.

Sample	Span	Width × thickness	Units / material
A			
B			
C			

02 Plan it on paper

Draw the setup with the loading point and a reference line for deflection.

My prediction, chosen feature or design limit

Compare beams under the same load

Task 2.4–2.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Load the samples one increment at a time. Measure center deflection at a common load. Unload and inspect. Compare which sample bent and recovered, which kept a bend, and which failed. Identify the limits of the comparison.

Sample	Common load and unit	Center deflection	Recovery after unloading
A			
B			
C			

04 Use the reading

Ch. 13 | Home Sweet Home | printed pp. 111-122
Ch. 14 | From the Ground Up | printed pp. 123-131
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Recommend a member for one part of your model. Cite a common-load measurement and describe the material response.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Compare two enclosures

Task 2.7 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch your insulation and thermometer position. Use identical warm-water containers and equal volumes. Mark the changed feature and keep sampling times equal.

Record starting temperature, volume, and elapsed time. Identify the one insulation feature to change.

Condition	Water volume	Start temp (°C)	Gap or wrap detail
Plain			
Insulated			
Resealed if used			

02 Plan it on paper

Draw the boundary of the box. Predict a heat-loss path and mark the thermometer location.

My prediction, chosen feature or design limit

Compare two enclosures

Task 2.7 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Prepare a baseline and insulated case. Record temperatures at equal intervals, using the same method. Compare temperature drops after the same interval. Seal one suspected gap and repeat if the starting conditions can be matched.

Elapsed time	Plain (°C)	Insulated (°C)	Resealed (°C)
Start			
Interval 1			
Interval 2			
Interval 3			
Final drop			

04 Use the reading

Ch. 15 | Building Green | printed pp. 133-142

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which enclosure lost less thermal energy under your test conditions? Support your inference with two measured temperature changes.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Draw a floor plan and elevation

Task 2.8 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Set a drawing scale and show a common corner. Draw walls, doors and the resident's trip in plan view. Beside it, add the elevation of one entrance.

Choose and label a scale. At 1/2 in = 1 ft, a 12-ft wall occupies 6 in. Convert one real dimension independently.

Feature	Real length and unit	Drawing length and unit	Scale
Wall			
Door			
Check in reverse			

02 Plan it on paper

Mark a common corner. Draw exterior walls, doors, and the two uses. Include a legend and units.

My prediction, chosen feature or design limit

Draw a floor plan and elevation

Task 2.8 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Add an elevation or detail showing a feature the floor plan cannot show. Mark the pedestrian route. Give the drawing to another crew. Ask them to recover a real dimension and trace the entrance without asking you.

Partner task	Recovered value or route	Clarification added
Find dimension		
Trace entry		

04 Use the reading

Ch. 16 | A Race for the Sun | printed pp. 143-151
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Show one scale conversion in both directions. What did the partner’s check make you change?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Turn the drawing into a building model

Task 2.9 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

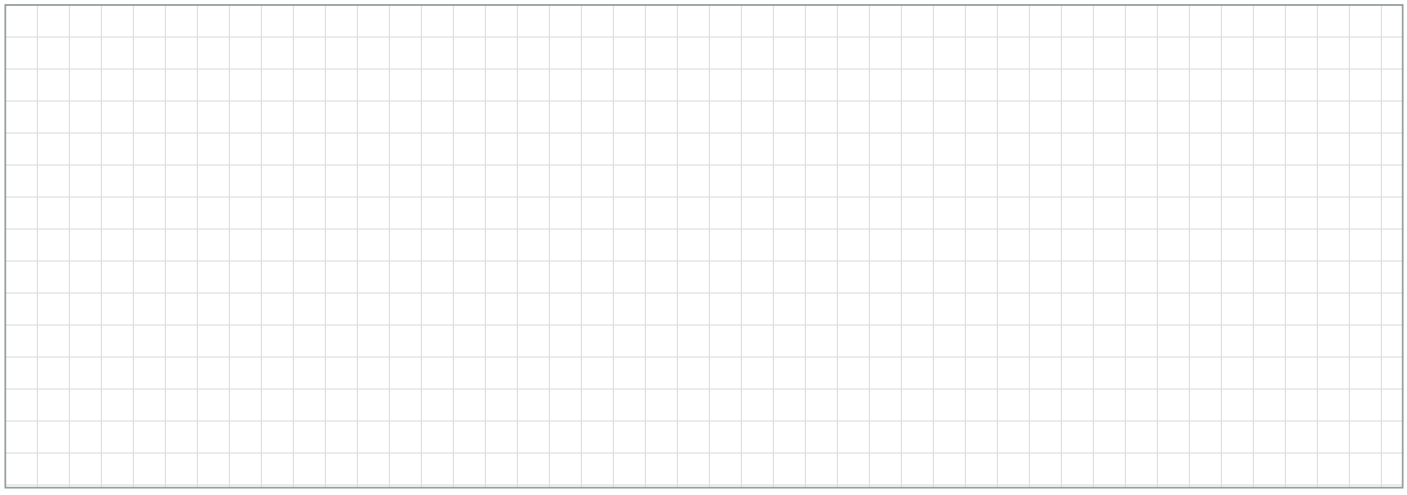
Mark an open side or removable roof and label both uses. Trace the route. Locate a construction detail supported by a load or insulation test.

Measure wall and opening sizes from the drawing. Mark the model dimensions on a cut list.

Cut-list item	Drawing dimension	Model target	Actual cut and units
Wall 1			
Wall 2			
Opening			
Divider			

02 Plan it on paper

Plan a removable roof or open side so reviewers can trace circulation. Identify one evidence-based construction detail.



My prediction, chosen feature or design limit

Turn the drawing into a building model

Task 2.9 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Make the shell, entrances, and interior divisions. Label housing access and the business/service. Add the structural or thermal detail you chose. Let another team follow the route and compare two model dimensions with the plan. Flag one mismatch for revision.

Peer check	Target	Model result	Mismatch or fix
Dimension 1			
Dimension 2			
Trace the route			

04 Use the reading

Ch. 11 | Bridging the Future | printed pp. 91-99
Ch. 15 | Building Green | printed pp. 133-142
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Which feature of the model represents a test result? Give the result and show where it changed the construction.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

Review a building as two users

Task 2.9 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Mark the altered feature on the as-built plan. Distinguish the resident’s route from the service user's route.

Measure the feature flagged at handover and compare with the planned value.

Reviewer role	Concern observed	Test or acceptance rule
Resident		
Service user		

02 Plan it on paper

Ask one reviewer to act as a resident and another as a service user. Write a testable concern from each.

My prediction, chosen feature or design limit

Review a building as two users

Task 2.9 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Revise a feature that addresses one concern. Mark the as-built change on the plan. Repeat the route or fit check. Present the final area, one physical test, and one trade-off.

Feature	Planned value	Before change	After retest and units
Dimension			
Route / access			
Physical test			

04 Use the reading

Ch. 10 | Redesigning America | printed pp. 83-87
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Defend one change to the client using a measurement, a drawing detail, and an idea from the reading.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.